Chapter 23

Vowel alternations in Dagaare number marking

Cassandra Caragine
University of Maryland

Dagaare (Mabia) has two numbers, singular and plural, but nouns form the singular and plural in a variety of ways depending on their declension class. In this paper, I provide novel data from Northern Dagaare to investigate the vowel alternations exhibited by three of the classes: -r marked singulars, -bɛ marked plurals, and -r(ɪ) marked plurals. Following previous work on Central Dagaare (Anttila & Bodomo 2009, 2019), I conclude that the vowel alternations in the -r singulars are in fact true suffixes. However, Anttila & Bodomo (2009, 2019) propose that the vowel alternations in the -r(ɪ) plurals are due to epenthetic vowels inserted to meet a word minimality requirement. I argue that this explanation does not hold in Northern Dagaare. Rather I propose that the singular suffix within the -r(ɪ) plural class is an empty mora. Additionally, I unify the -bɛ and -mɛ plural classes in Northern Dagaare and provide a preliminary analysis to explain the compensatory lengthening that occurs in the former but not the latter.

1 Introduction

The Dagaare (Mabia)[1] language is made up of a dialect continuum which stretches from Northern Ghana to southern Burkina Faso. In this paper, I present novel data from the Nandome dialect of Northern Dagaare, which exhibits vowel alternations between several singular and plural nouns. Anttila & Bodomo (2009, 2019) found that similar patterns in Central Dagaare are driven by metrical foot structure, minimal prosodic word requirements and other language general

[1] Dagaare is often also called Dagaari, Dagara, Waale, and Birifor. The Mabia language family is sometimes referred to as Gur.

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constraints. I investigate whether this analysis can be extended to data from Northern Dagaare. I argue that despite their success capturing the variation in Central Dagaare, this question in the larger Dagaare language is far from settled. While the Anttila and Bodomo analysis makes the correct predictions for the -r singulars, it does not accurately explain the variation seen in the -r(ɪ) plurals and it does not address vowel length alternations in the -bɛ plurals.

I maintain a simple affixation account of the -r singulars put forth by Anttila & Bodomo (2009), but I propose that within the -r(ɪ) plural declension class, the singulars receive a suffix consisting only of an empty mora. This mora is then filled by linking it to the features of the preceding vowel, resulting in vowel lengthening. Then language general constraints adopted from Anttila & Bodomo (2009) sometimes cause diphthongization. In the -bɛ plurals, I argue that there is compensatory lengthening of the stem vowel in the plural when the stem final consonant is deleted. When the final consonant is a nasal it coalesces with the [b] in the suffix to form an [m]. This [m] is ambisyllabic and thus no compensatory lengthening is triggered.

The paper is organized as follows. Section 2 introduces the Northern Dagaare dialect and its relevant linguistic features, Section 3 presents data of number marking, Section 4 works towards an analysis, and Section 5 concludes.

2 Northern Dagaare

2.1 Language background

Dagaare is an understudied and under-documented Mabia language (Niger-Congo) spoken in northwestern Ghana and southern Burkina Faso by between 1.5 and 2 million speakers (Ali et al. 2021, Lewis 2009). The Dagaare language is made up of a dialect continuum that, roughly speaking, can be divided into four main dialect groups – Northern, Southern, Western and Central – which show substantial variation (Bodomo 1997, Angsongna & Akinbo 2020, Ali et al. 2021). The map in Figure 1 shows the locations in which the dialect groups are spoken. This paper focuses on Nandome Dagaare, an under-documented sub-dialect of Northern Dagaare spoken in and around Nandom, Ghana.

A team of researchers at Georgetown University[2] and the University of Maryland[3] has spent over a year working with a native speaker to document and analyze this understudied dialect. The speaker is a male (27), who grew up in

[2] Michael Obiri-Yeboah, Sophie Migacz, and Jack Pruett

[3] Cassandra Caragine

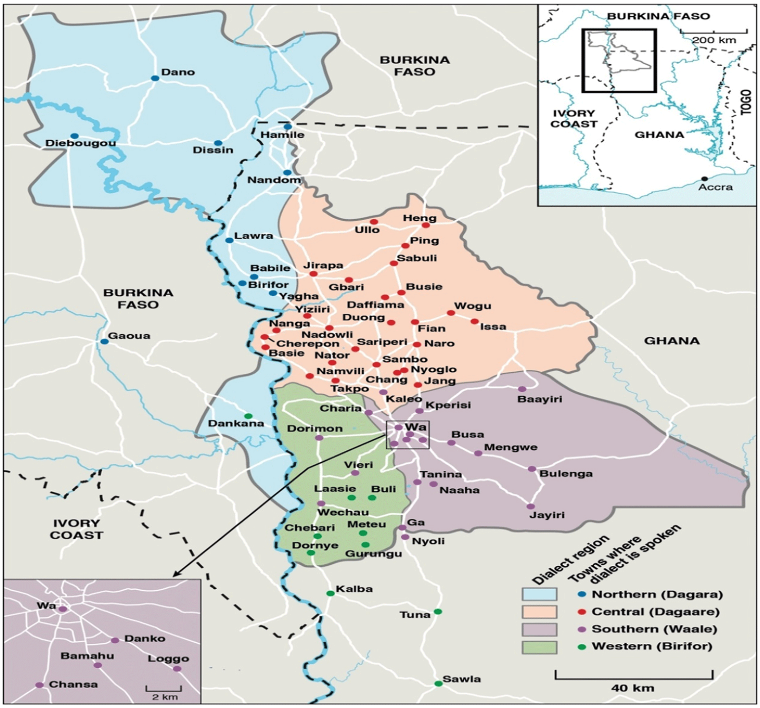

Figure 1: Map of the area where Dagaare is spoken (Angsongna & Akinbo 2020)

Kumasi, Ghana and speaks the Nandome dialect of Dagaare. Additionally, he speaks Twi, English, French, and Spanish. All interview sessions were recorded (about 40 hours) with a Zoom H4n Pro handheld recorder. Data was transcribed and uploaded to Twisted Tongues database management software. Researchers used Praat (Boersma & Weenink 2023) for confirmation and correction of initial transcriptions. This resulted in 68 singular-plural pairs in Northern Dagaare.

2.2 Phonological background

Northern Dagaare (ND) has a nine-vowel system consisting of four +/-ATR (Advanced Tongue Root) pairs and one low central vowel, Table 1. The low central vowel /a/ is likely under-specified for ATR; evidence for this is presented later in this section. Vowels also contrast in length ([nà] 'this' v. [nàà] 'chief'), tone ([báá] 'dog' v. [bàà] 'river'), and nasality ([sàà] 'rain' v. [sã̀ã̀] 'father').

Additionally, ND exhibits ATR vowel harmony. In the nominal domain, this process is progressive. The low central vowel /a/ is invisible to this process; it

Table 1: Dagaare Vowel Chart

	Front		Back	
	+ATR	-ATR	+ATR	-ATR
High	i	ɪ	u	ʊ
Mid	e	ɛ	o	ɔ
Low		a		

does not undergo, trigger, or block vowel harmony. Thus, it is likely that /a/ is underspecified for ATR. For further discussion of vowel harmony in Dagaare see Bodomo (1997) and Ali et al. (2021).

Northern Dagaare has a (C)V(V)(C) syllable structure. It allows optional onsets and optional codas as well as short vowels, long vowels, and diphthongs. Complex onsets are currently unattested, and there is only one word in our data set which contains a complex coda: [tʊ̀βr] 'ears'. Syllable weight and moraic structure will be considered in Section 4.3.

3 Number marking

The Dagaare nominal system has two numbers: singular and plural. Dagaare is noted to have an 'inverse' or 'polarity' system, in which a single morpheme can act as either a singular or plural marker (Grimm 2012, 2021, Anttila & Bodomo 2009). As Table 2 shows, -r can mark either singulars or plurals depending on the declension class of the noun.

Table 2: Polarity Marking

Singular	Plural	Gloss
nóbíè	nóbí-r	'finger'
jí-r	jíé	'house'

However, -r is not the only number marker. There are several ways that number can be expressed on the noun depending on its declension class. The declension classes of Dagaare have been categorized in several ways. I use the six classes shown in Table 3, but the exact categorization is tangential to the current paper.

Additionally, the labels used for each class have not be standardized and are provided for referential simplicity only. As noted in Section 2.2, vowel harmony is an active process in the language. The citation forms of each of the suffixes is given in the -ATR form as this is what occurs with stems whose only vowel is /a/, which is underspecified for ATR. The specific rules governing which number markers are used for a particular noun are outside the scope of this paper. For further discussion of this topic see Pruett (n.d.).

Table 3: ND Declension Classes

	Singular	Plural	Gloss
-r singulars	jí-r	jí-é	'house'
-r(ɪ) plurals	bàà	bàà-r	'river'
-bɛ plurals	dɛ́b̚	dɛ̀ɛ̀-βɛ̀	'man'
l nominals	jéèpùlè	jéépùlì	'sister'
-mɪnɛ plurals	sã̀ã̀	sã̀ã̀-mìnɛ́	'father'
irregulars	wáàb	wíír	'snake'

The declension classes and a representative example for each are included in 3. The first three of these classes (-r singulars, -r(ɪ) plurals, and -bɛ plurals) exhibit some sort of vowel alternation. Sections 3.1-3.3 will lay out the patterns seen in these noun classes in Northern Dagaare and compare them to their correspondents in Central Dagaare (CD).

3.1 -r singulars

The -r singular declension class is one of the classes exhibiting polarity marking with -r(ɪ). As can be seen in Table 4, this class overtly marks both the singular and plural with a suffix. An -r is suffixed in the singular and -ɛ is suffixed in the plural. The -ɛ plural suffix undergoes ATR vowel harmony to agree with the stem. When the -ɛ suffix results in vowel hiatus, there is no rearticulation. When the qualities are different for the two vowels, the phonetic yield is a diphthong, and when the two vowels have the same quality (eg. when -ɛ is added to an [ɛ] or [e] final stem), a long vowel is produced.

The left 2 columns of Table 4 provide novel singular/plural pairs in the -r singular declension class of Northern Dagaare. The right two columns provide the corresponding word in Central Dagaare.[4] Unless otherwise noted, here and throughout the paper, the Central Dagaare data come from Ali et al. (2021).

Table 4: -r singulars in Northern Dagaare and corresponding nouns in Central Dagaare. Central Dagaare data from Ali et al. (2021)

Northern Dagaare			Central Dagaare	
Singular	Plural	Gloss	Singular	Plural
bàpɛ́-r	bàpɛ́-ɛ́	‘male friend’	bábí-rì	*babi-e (‘best friend’)
g͡bɛ́-r	g͡bɛ́-ɛ́	‘leg’	*g͡bɛ́-rì	*g͡bɛ-ɛ
tʊ̋β-r	tʊ̀β-ɛ́	‘ear’	tòò-rí	*tobo
jí-r	jí-é	‘house’	jí-rì	*ji-e
dɛ̀-r	dɛ̀-ɛ̀	‘stair’	–	–
bȕbí-r	bȕbì-é	‘seed’	bʊ̋mbí-rì	*bombi-e
mʊ̀ndɛ́-r	mʊ̀ndɛ́-ɛ́	‘spider’	bádɛ́-rí	*badɛr-rɪ
màrbí-r	màrbí-è	‘star’	ŋ͡máríbí-rì	*ŋ͡marebi-e

As can be seen in Table 4, the the singular -r suffix in Northern Dagaare always corresponds to a -rɪ suffix in Central Dagaare. ‘Ear’ and ‘spider’ appear to be irregular plurals in CD, but the remaining CD plurals are marked with a corresponding suffix to that used in ND.

3.2 -r(ɪ) plurals

The most common of the declension classes adds -r or -rɪ in the plural (18/68 noun pairs). Exactly what conditions the final [ɪ] is still unknown.[5] Other dialects

[4] In the original sources standard Dagaare orthography was used instead of IPA. For consistency and ease of comparison, the orthography has been converted to IPA. There is generally a one-to-one correspondence between the orthography and IPA. The only exceptions to this are <e> which can be either [e] or [ɪ] and <o> which can be either [o] or [ʊ]. I used whichever IPA symbol agreed in ATR with the rest of the word. When the context was ambiguous, I selected the symbol which more closely resembled the ND pronunciation. Tones are not recoverable from the orthography, so they have been omitted. These reconstructed examples have been marked with an asterisk (*). A dash (–) indicates that a translation could not be found for this word. When there is a semantic difference between the ND and CD words, the CD meaning is indicated in parenthesis.

[5] There may also be some synchronic variation. When used in a sentence context rather than in isolation, those forms which still “retain” the final [ɪ] may reduce/delete the final vowel [tí-ɾí] ~[tíɾ] ‘tree’. This can alternate within a single speaker from one utterance to another. It is possible that since the -r is the only information needed to distinguish the singular from the plural, Northern Dagaare may be losing this final vowel in more common nouns. Further research is required to determine exactly where and why the final [ɪ] is sometimes dropped.

seem to only have -rɪ (Anttila & Bodomo 2009, 2019, Bodomo & Marfo 2007) – see Tables 5-6 below for examples in CD. Interestingly, as was seen in Section 3.1, the [ɪ] has universally been lost when it marks the singular of the -r singular declension class even though their correlate in other dialecs always has a final [ɪ]. Table 5 shows some examples of plurals which are formed by the addition of -r(ɪ).

Table 5: -r(ɪ) plurals in Northern Dagaare and corresponding nouns in Central Dagaare

Northern Dagaare			Central Dagaare	
Singular	Plural	Gloss	Singular	Plural
báá	báá-r	'dog'	báá	*báá-rí (Bodomo 1997)
bàà	bàà-r	'river'	bàá	*bà-rí (Bodomo 1997)
dáá	dáá-r	'stick'	dàá	*dà-rí (Bodomo 1997)
nàfáʊ̀	nàfáʊ̀-ɾí	'shoe'	*nɔɔtɪ-rʊʊ	*nɔɔtɪ-ɛ
gàdò	gàdò-ɾí	'bed'	gádó	*gado-ri
mʊ̀ɔ́	mʊ̀ɔ́-r	'grass'	mɔ́-rí	*mɔ-ɛ

As seen in Table 5, in this declension class the singulars are unmarked and the plurals are marked by either -r or -ɾí.[6] When the plural marker has a final [ɪ], there is a general allophonic process in ND in which /r/ becomes [ɾ] in syllable onsets.

As can be seen in Table 6, some nouns in the -r(ɪ) plural declension class show variations in the vowels between the singular and plural forms. The singular nouns in ND have a vowel which does not appear in the plural. In some words, like [núú], the extra vowel in the singular is the same as the vowel in the plural stem [nú], but in others, for example [tíɛ̀], the vowel which appears in the singular is different from the vowel in the plural stem [tí]. There are also vowel alternations in the CD forms, but of a different sort. 'Hand' has an extra vowel in the plural rather than in the singular. Also there is a vowel quality difference between the singulars and plurals of 'tree', 'thief', and 'goat'. Another noteworthy fact is that the words 'eye', 'finger', and 'toe' are in different declension classes in the two dialects; while they are in the -rɪ plural class in ND, they are in the -r singular class in CD.

[6] Note that in Tables 5-6 all stems which take the -r(ɪ) plural end in vowels. For discussion of how this may impact declension class, see Pruett (n.d.)

Table 6: -r(í) plurals in Northern Dagaare and corresponding nouns in Central Dagaare

Northern Dagaare			Central Dagaare	
Singular	Plural	Gloss	Singular	Plural
nímíè	nímí-r	‘eye’	nímí-rì	*nimi-e
nóbíè	nóbí-r	‘finger’	núbì-rì	*nubi-e/*nubii-ri
g͡bébíè	g͡bébí-r	‘toe’	gbɛ́bì-rí	*gbɛbi-e
tíɛ̀	tí-ɾí	‘tree’	tìɛ́	tìì-rí (Bodomo 1997)
ɲɑ̃̀ɲúé	ɲɑ̃̀ɲú-r	‘thief’	nàɲɲìgé	*nanɲigi-ri
bùó	bó-r	‘goat’	bʊ́ɔ́	bʊ́ʊ́-rí
núú	nú-r	‘hand’	nû	*nuu-ri

3.3 -bɛ plurals

In the -bɛ plural declension class, the singular is unmarked and the plural is marked by -bɛ. The -bɛ plurals surface in two ways: -βɛ and -mɛ. We see -βɛ appear when the suffix is added to plosives final stems and -mɛ appears when it is added to nasal final stems. These appear to be the only stem types which take the -βɛ plural suffix.

As can be seen in Table 7, when -bɛ is added to a plosive-final stem, the plosive deletes. Additionally, there is a language general spirantization process which turns [b] into [β] between most sonorants.

Table 7: -bɛ plurals in Northern Dagaare and corresponding nouns in Central Dagaare. Central Dagaare data from Ali et al. (2021).

Northern Dagaare			Central Dagaare	
Singular	Plural	Gloss	Singular	Plural
dɛ́b̚	dɛ̀ɛ̀-βɛ̀	‘man’	dɔ́ɔ́	*dɔ-bɔ
jɛ̀b̚	jɛ̀ɛ̀-βɛ́	‘sibling’	jɔ́ɔ̀	jɔɔmine/*jɔɔre
pɔ́g̚	pɔ́ɔ́-βɛ̀	‘woman’	pɔ́g-ɔ́	pɔ́g-bɔ́

As shown in Table 7, there are vowel length alternations between the ND singulars and plurals in the -bɛ class.[7] The stem vowel is always short in the

[7]One may note that all of the words in Table 7 contain -ATR mid vowels and end in voiced consonants, but I assume this is a coincidence of the small dataset.

singular but long in the plural. However, this alternation is not universal in the class. There is a group of nouns in the -bɛ plural declension class which does not show any vowel alternations (Table 8).

Table 8: -mɛ plurals in Northern Dagaare and corresponding nouns in Central Dagaare. Central Dagaare data from Ali et al. 2021.

Northern Dagaare			Central Dagaare	
Singular	Plural	Gloss	Singular	Plural
nɛ́n	nɛ́mɛ́	‘meat’	nɛ́nì	nɛ́mɛ̀ (Bodomo 1997)
bʊ́n	bʊ̀mɛ́	‘thing’	bʊ́ní	*bʊma
pán	pàmɛ́	‘door’	pání	*pama (‘doorway’)
dàt͡ʃín	dàt͡ʃímé	‘wall’	dànʧíní	*dant͡ʃime
k͡pàk͡pán	k͡pàk͡pámɛ́	‘arm’	kpáŋkpání	*kpaŋkpama
zʊ́m	zʊ́mɛ̀	‘fish’	zʊ́mmʊ́	*zʊma
pìpán	pìpámé	‘rock’	píírí	*pie
gán	gámɛ́	‘book’	gánì	gámá
sáán	sáámɛ́	‘visitor’	sáánà	saama (Bodomo 1997)

As alluded to earlier, all of the nouns which end in a nasal consonant in the singular receive the -mɛ suffix in the plural. The -mɛ plural has historically been treated as a separate class (Bodomo & Marfo 2007). However, I argue it is an allomorph of the –bɛ plural when it is added to nasal final stems. When -bɛ́ is added to a nasal-final stem, the nasal and the voiced bilabial plosive [b] coalesce to form a bilabial nasal [m]. The nouns which have a nasal final consonant and take the -bɛ plural suffix do not undergo any sort of vowel alternation. The short vowels in the singular remain short when the plural suffix is added.

4 Towards an analysis

As shown in Section 3, several Northern Dagaare nouns exhibit alternations in vowel length and diphthongization between the singular and plural. Seeing data like this in a dialect of Central Dagaare spoken around the towns Jirapa and Ullo, Anttila & Bodomo (2009) formed an account for number marking based on minimal prosodic word requirements. In this section, I attempt to apply this account to our Northern Dagaare data, and analyze the instances in which it can capture the variation in our data and the places in which it fails. When their account cannot accurately capture the ND facts, I put forth an alternative analysis. Like Anttila & Bodomo (2009), throughout the course of this analysis, I take ATR harmony for granted and leave out tone for simplicity.

The section begins with a discussion of -r singulars in 4.1 which function identically to those in Central Dagaare. In 4.2 I show that the word minimallity requirement in CD may not be active in ND and propose an alternative analysis. Section 4.3 discusses the -bɛ plurals, not analyzed in Anttila & Bodomo (2009, 2019), and provides a potential direction for analysis of the class.

4.1 -r singulars

As shown earlier, in the -r singular declension class, nouns receive an -r suffix in the singular and an -ɛ suffix in the plural. The plural suffix often results in a dipthong or a long mid vowel. As motivation for a processes which will be discussed later, Anttila & Bodomo (2009) posit a constraint against derived long mid vowels. However, since -ɛ is a suffix and not an epenthesized vowel, the long mid vowel is allowed. They justify this in OT by having an IDENT(V) constraint ranked higher than *VV(mid). These constraints, and the widespread MAX constraint are defined below in (1).

(1) a. MAX: Assign one violation mark for each segment present in the input which does not have a correspondent in the output.
    b. *VV(mid): Assign one violation mark for each long mid vowel.
    c. IDENT(V): For a vowel in the input V_i which has a corresponding output vowel V_o, assign one violation for each feature value which is not shared between V_i and V_o.

Using these constraints, the derivation of the Northern Dagaare word 'legs' can be seen in Table 9 below. Thus far, this analysis seems to directly capture what we see in Northern Dagaare. The plural suffix -ɛ is added to the stem, resulting in a long mid vowel. Additionally, as the /ɛ/ is in the underlying form, the long mid vowel is allowed to surface.

4.2 -r(ɪ) plurals

In the -r(ɪ) plural declension class, the plural receives either the suffix -r or -rɪ while the singular is unmarked. As described in Section 3.2, some nouns have a long vowel or diphthong in the plural which is not present in the singular.

One could posit that one vowel in a VV sequence is deleted when the plural morpheme -r(ɪ) is added, and the lack of vowel shortening in Table 5 (repeated

Table 9: Tableau deriving [g͡bɛ́ɛ́] 'legs' in ND

/g͡bɛ-ɛ/	Ident(V)	Max	*VV(mid)
☞ a. g͡bɛɛ			*
b. g͡bɪɛ	*!		
c. g͡bɛɪ	*!		
d. g͡bɛ		*!	

below in Table 10 for convenience) is due to a preference for low vowels, as we do not see any low vowels deleting. However, [nàfáʊ̀] ends in a high vowel which seems like it would be a good candidate to delete, yet it remains [nàfáʊ̀ɾí] in the plural. Thus, I follow A&B in assuming that this is an epenthesis process and the fact that only low vowels are long in the plural is merely an idiosyncrasy of this limited data set. This conclusion is further supported by considering the unpredictability of which vowel appears in the plural form. For example, when the singular form contains a diphthong, [tíɛ̀] and [bùó], sometimes the first vowel appears in the plural [tíɾí] and sometimes the second does [bór]. A deletion account could not explain this alternation.

Table 10: -r(ɪ) plurals

Singular	Plural	Gloss
báá	báá-r	'dog'
bàà	bàà-r	'river'
dàà	dáá-r	'stick'
nàfáʊ̀	nàfáʊ̀-ɾí	'shoe'
gàdò	gàdò-ɾí	'bed'
nímíè	nímí-r	'eye'
nóbíè	nóbí-r	'finger'
g͡bébíè	g͡bébí-r	'toe'
tíɛ̀	tí-ɾí	'tree'
ɲã̀ɲúé	ɲã̀ɲú-r	'thief'
bùó	bó-r	'goat'
núú	nú-r	'hand'

Anttila & Bodomo (2009) state that in Central Dagaare, when a stem is monomoraic, a vowel is epenthesized in the singular to create a fully formed bimoraic foot. They state that this is a constraint seen across the language; all phonological words must minimally contain a fully formed prosodic foot. To satisfy this constraint, a vowel is added to the end of monomoraic stems. This vowel matches the quality of the previous vowel. However, as can be seen in Table 10, the vowels do not always match. Anttila and Bodomo argue that this is due to language general constraints against word final high vowels and long mid vowels (breifly mentioned above).

The following tableau are modeled after Anttila & Bodomo 2009, but the data are from Northern Dagaare. The relevant constraints they describe are as follows:

(2) a. Dep(V): Assign one violation mark for each vowel in the output which does not have a correspondent in the input.
b. FtBin: Assign one violation mark for each foot which is not minimally bimoraic.
c. *I]: Assign one violation mark for each word-final [+high] vowel.

Table 11: Tableau deriving [tɪɛ] 'tree' in ND

/tɪ/	FtBin	Dep(V)	*I]
a. tɪ	*!		*
b. tɪɪ		*	*!
☞ c. tɪɛ		*	

Table 11 shows the derivation of [tɪ́ɛ̀] 'tree'. The stem is monomoraic, so a vowel is epenthesized to satisfy FtBin. You would normally epenthesize a vowel with the same features as the stem vowel. However, to satisfy the constraint against word final high vowels, a mid vowel is added instead.

But, as can be seen in Table 10, there are several words which end in high vowels. Similar to the analysis in 4.1 of the *VV(mid) constraint, the constraint against word final high vowels only affects "derived vowels" – i.e., those which do not have a corespondent in the UR. This is again attained by a higher ranked Ident(V) which prevents vowel lowering if the violating high vowel has a correspondent in the input. A simple case of this can be seen in Table 12.
Despite violating *I], the [i] remains in the output because changing it would violate the higher ranked Ident(V).

As noted in Section 3.2, the Central Dagaare plural suffix, -rɪ, often appears without the vowel in Northern Dagaare. This is important because while the plural marker straight forwardly satisfies the FtBin constraint in Central Dagaare,

Table 12: Tableau deriving [gadoɾi] 'bed' in Northern Dagaare

/gado-ɾi/	IDENT(V)	DEP(V)	*I]
☞ a. gadoɾi			*
b. gadore	*!		
c. gadoɾie		*!	

in Northern Dagaare it is necessary to say that /r/ is moraic so as to satisfy the foot minimality requirement and not trigger vowel epenthesis in the plural. The moraic status of coda consonants will be discussed further in Section 4.3.

Table 13: Tableau deriving [buo] 'goat' in Northern Dagaare

/bo/	FTBIN	DEP(V)	*I]	*VV(MID)
a. bo	*!			
b. boo		*		*!
☞ c. buo		*		
d. bou		*	*!	

In Table 13, we see the derivation of [bùó] 'goat'. Again, the stem is monomoraic so a vowel must be epenthesized. The constraint against long mid vowels prevents the normal epenthesis of [o] and instead a high vowel is epenthesized. However, due to the constraint against word final high vowels, the epenthesized vowel cannot occur word finally as it has in the other examples. Rather, the high vowel [u] is epenthesized before the stem vowel.[8]

This accounts well for Bodomo and Anttila's data as well as the Northern Dagaare data discussed above, but when attempting to expand this theory to a larger set of ND nouns, it does not capture all the variation. The word minimality account does not explain the epenthetic vowel in the singulars of nouns with multi-moraic stems (Table 14).

As the stems in Table 14 are already, at minimum, bimoraic, Anttila and Bodomo's account does not provide an explanation for the epenthetic vowels in the singular. In fact, as seen in Table 15, their constraints would make the wrong prediction.

[8]One may note that there is nothing particular indicating where the epenthesized vowel appears or what quality it should take. This is a shortcoming of the original analysis. Since I will soon argue that this analysis does not work for ND and propose an alternative, I will not try to plug this hole. However, future work on CD would benefit from a more complete analysis of the nature and location of the epenthesized vowel.

Table 14: Multimoraic Stems of ND -ɾɪ plurals

Stem	Singular	Plural	Gloss
nóbì	nóbíé	nóbí-r	‘finger’
nímí	nímíè	nímí-r	‘eye’
g͡bébí	g͡bébíè	g͡bébí-r	‘toe’
tèɾét͡ʃì	tèɾét͡ʃìé	tèɾét͡ʃì-ɾí	‘train’
ɲȁa̋ɲù	ɲȁa̋ɲùé	ɲȁa̋ɲù-ɾí	‘thief’

Table 15: Tableau of the failed derivation of [nóbíé] ‘finger’ in ND

/nobi/	FtBin	Dep(V)	*I]	*VV(mid)
💣 a. nobi			*	
☹ b. nobie		*!		
c. nobii		*!	*	

Finally, the prosodic account put forth by Anttila and Bodomo is dependent on the fact that Dagaare requires a fully formed foot for all prosodic words. They acknowledge that there are a few exceptions to this rule for Central Dagaare, specifically [bâ] ‘father’, [mǎ] ‘mother’, [nû] ‘hand’, [zû] ‘head’, all of which are very high frequency words. However, deficient feet seem to be much more common in Northern Dagaare. In addition to those mentioned for Central Dagaare, the Northern dialect also has [nú] ‘face,’ as well as many verbs including but not limited to [sʊ̀] ‘bathe’, [ɲʊ̀] ‘drink’, and [tʃɛ́] ‘slice.’

There may be a concern that the verbs do not in and of themselves form full prosodic units as they typically exist as part of a verbal paradigm including auxiliaries, tense markers, and object pronouns. However, we see that verbs can stand alone in imperatives (3).

(3) wá
‘come’

Word minimality does not seem to be as active of a constraint in Northern Dagaare as it was in Central Dagaare. Thus, using it as a motivation for vowel epenthesis in nominals is less obvious. Additionally, epenthesis occurs in some words which already satisfy the proposed word minimality.

Rather, following work on quantity-manipulating morphology (Fujimura 1999, Trommer & Zimmermann 2014), I propose that there is an empty mora suffixed to all singular nouns in the -rɪ plural class. This mora is then filled by linking to the adjacent vowel. I will generally assume an autosegmental feature geometry such as those proposed in Clements (1985), but the specifics of the system are not central to the analysis. Having a mora be the affix which spells out singular number requires that all nouns in this class will have an "epenthetic" vowel due to Max-μ. This must be ranked over 1-to-1, a constraint against lengthening/diphthongization of the vowel.

(4) a. Max-μ: assign one violation mark for each mora in the input which is not filled in the output.

b. 1-to-1: Assign one violation mark for each segment that is linked to more than one prosodic node.

The same constraints proposed by Anttila & Bodomo 2009 can be used to account for the instances in which the vowel qualities do not match. If after linking your final vowel to the empty mora, the final vowel is high, the vowel is lowered to satisfy *I] (Table 16). If after linking, there is a long mid vowel, in violation of *VV(mid), the first element of the vowel sequence is raised (Table 17).

Table 16: Tableau illustrating high vowel lowering in [nóbíé] 'finger' in ND

/nobi$^{\mu - \mu}$/	Max-μ	*I]	1-to-1
a. nobi$^{\mu}$	*!	*	
☞ b. nobi$^{\mu}$e$^{\mu}$			
c. nobi$^{\mu\mu}$		*!	*

Table 17: Tableau illustrating mid vowel raising in [buo] 'goat' in ND

/bo$^{\mu - \mu}$/	Max-μ	*VV(mid)	1-to-1
a. bo$^{\mu}$	*!		
☞ b. bu$^{\mu}$o$^{\mu}$			
c. bo$^{\mu\mu}$		*!	*

As shown in Table 18, not all singular nouns in this class undergo vowel lengthening. There is no lengthening in stems which already contain a long vowel. This can easily be explained, as there are no "superlong" vowels in Dagaare.

Under this analysis, in the -r(ɪ) plural class, nouns in the singular – so long as they do not already end in a long vowel – will always undergo vowel lengthening or diphthongization due to the added mora. This contrasts with Anttila and

Table 18: Tableau showing no lengthening of [daa] 'stick' in ND

/da$^{\mu\mu}$-μ/	*SuperLong	Max-μ
a. da$^{\mu\mu\mu}$	*!	
☞ b. da$^{\mu\mu}$		*

Bodomo's word minimallity analysis which would only predict vowel lengthening/diphthongization on stems that are less than a fully formed foot.

It should be noted that this analysis would not correctly predict the ND singulars [gàdò] 'bed' and [núú] 'hand' . Mora affixation accompanied by mid vowel raising would predict [gàdùò]. Additionally, the constraint *I] would predict high vowel lowering in 'hand' to produce [núó]. However, this would also be incorrectly predicted under Anttila and Bodomo's account.[9]

4.3 -bɛ́ plurals

Anttila and Bodomo did not address the -bɛ plurals in their work on Central Dagaare.[10] As shown in Table 7 the -bɛ́ plurals in ND have a short vowel in the singular and a long vowel in the plural. Logically, this could be due to shortening in the presence of a coda or lengthening with the deletion of an underlying coda. However, there are many words with long vowels and codas (all of the -r plurals with long vowels as well as several other words including [wáàb] 'snake', [bɛ́ɛ́n] 'one', and [lɛ̀ɛ̀m] 'taste') so it seems unlikely that the vowel is shortened in the presence of a coda.

Therefore, I conclude that the -bɛ plurals are undergoing a process of compensatory lengthening, in which one segment is lengthened to compensate for the loss of another. For example, /pɔ̀g-bɛ́/ 'women' becomes [pɔ́ɔ́βɛ̀]. Here the vowel lengthens to compensate for the loss of the [g]. Additionally, there is a language general process of /b/ spirantization between sonorants accounted for by a Spirant constraint. Compensatory lengthening cases such as these are frequently analyzed with a moraic account (Hayes 1989, Shaw 2008). A moraic coda, when deleted, leads to an empty mora which is then filled by lengthening of the preceding vowel. I consider this deletion to be triggered by a *CC constraint specified to ban adjacent [-sonorant] consonants. I propose this holds across the language, as we never see two adjacent obstruents. Additionally, the deletion of the first

[9] In CD [nû] 'hand' is also a difficult case which would not be predicted under Anttila and Bodomo's word minimality account.

[10] This in unsurprising since, as can be seen in Tables 7 and 8, it is likely that the -bɛ plurals do not actually form a unified class in CD.

rather then the second obstruent is due to the ranking of MAX$_{onset}$ over a general MAX constraint.

(5) a. *CC: Assign one violation mark for each consonant cluster in which both segments are [-sonorant].
b. MAX$_{onset}$: Assign one violation mark for each onset in the input which does not have a corespondent in the output.
c. SPIRANT: Assign one violation mark for each [-son] segment between two [+son] segments.

Table 19: Tableau deriving [pɔ̀ɔ̀βɛ́] 'women'

/pɔ$^{\mu}$g$^{\mu}$-bɛ/	*CC	MAX$_{onset}$	MAX-μ	SPIRANT	1-TO-1	MAX
a. pɔ$^{\mu}$g$^{\mu}$bɛ	*!					
b. pɔ$^{\mu\mu}$bɛ				*!	*	*
c. pɔ$^{\mu\mu}$ɣɛ		*!			*	*
☞ d. pɔ$^{\mu\mu}$βɛ					*	*
e. pɔ$^{\mu}$βɛ			*!			*

In Table 19 *CC in conjunction with MAX$_{onset}$ causes the deletion of the coda [g]. In compensation for this, triggered by MAX-μ, the vowel lengthens to fill the empty mora.

However, as seen in Section 3.3, when the -bɛ plural suffix attaches to nasal final roots, the /b/ and the final nasal coalesce into an [m] – [pán] 'door', [pàmɛ́] 'doors.' Thus, UNIFORMITY, a constraint preventing coalescence, must be ranked below *NC, which triggers coalescence. Additionally, since the [nasal] and [labial] features both surface on the resulting segment when the [n] and [b] coalesce, I introduce MAX$_{[nasal]}$ and MAX$_{[lab]}$.

(6) a. UNIFORMITY: assign one violation mark for each element in the output that has more than one correspondent in the input.
b. MAX$_{[\text{nasal}]}$: assign one violation mark for each [nasal] feature in the input that has no correspondent in the output.
c. MAX$_{[\text{lab}]}$: assign one violation mark for each [labial] feature in the input that has no correspondent in the output.
d. *NC: assign one violation mark for each nasal-consonant sequence.

Table 20: Tableau illustrating the failed derivation of [pàmɛ́] 'door'

/pa$^{\mu}$n$_{1}^{\mu}$-b$_{2}$ɛ/	Max$_{[\text{LAB}]}$	Max$_{onset}$	*NC	Max$_{[\text{NASAL}]}$	Max-μ	1-to-1	Max	Uni
a. pa$^{\mu}$n$_{1}^{\mu}$.b$_{2}$ɛ			*!					
b. pa$^{\mu\mu}$.b$_{2}$ɛ				*!		*	*	
c. pa$^{\mu}$.b$_{2}$ɛ				*!	*		*	
d. pa$^{\mu\mu}$.n$_{1}$ɛ	*!	*				*	*	
☹ e. pa$^{\mu}$.m$_{12}$ɛ					*!			*
💣 f. pa$^{\mu\mu}$.m$_{12}$ɛ						*		*
g. pa$^{\mu\mu}$.n$_{12}$ɛ	*!					*		*

When a nasal and [b] coalesce, there is no change in vowel length. As can be seen in Table 20, the same ranking which correctly captured compensatory lengthening in obstruent final roots fails to correctly predict the behavior of nasal final roots. If we assume Onset Maximization (Ito 1989, Selkirk 1982) the [m] would be the onset of the second syllable, not the coda of the first. Therefore it would not be able to hold a mora, and [paamɛ] would incorrectly win to satisfy Max-μ. In some way, the retention of the nasal feature from the underlying coda must "count" as filling a second mora. One potential explanation for this would be that the resulting [m] is ambisyllabic; this way it could satisfy Onset Maximization but still be able to hold a mora. Table 21 shows what this could look like in an OT tableau.

Table 21: Tableau deriving [pàmɛ́] 'door'

/pan$_{1}$-b$_{2}$ɛ/	*NC	Max-μ	1-to-1	Max	Uni
a. pa$^{\mu}$.m$_{12}$ɛ		*!			*
☞ b. pa$^{\mu}$(m$_{12}^{\mu}$)ɛ					*
c. pa$^{\mu\mu}$.m$_{12}$ɛ			*		*

In Table 21, Candidates (a) and (b) are segmentally identical, but in (b) one of the mora in the first syllable is filled by some of the feature information of the underlying /n/. If the [m] is ambisyllabic, filling both the coda of the first syllable and the onset of the second, we may expect to see some phonetic evidence. I leave investigation of this possibility to future research.

5 Conclusion

Anttila & Bodomo (2009) formulated an analysis of vowel alternations in Central Dagaare. Their analysis of the -r singulars seems to map directly to the -r singulars in Northern Dagaare. However, they argue that for Central Dagaare, vowels are epenthesized in the singular to fulfill a word minimality requirement for bimoraic feet. But, I have shown that in Northern Dagaare some words have epenthesized vowels even when they already contain bimoraic feet. Instead, I propose that the singular marker for the -r(ɪ) plural class is an empty mora. This mora suffix is then filled by linking to the adjacent vowel, and language general constraints formulated by Anttila and Bodomo modify the vowels to form the correct surface forms.

Additionally, I put forth a preliminary account for vowel length alternations in the -βɛ and -mɛ plurals, here analyzed as a single class, explaining why compensatory lengthening occurs in the former but not the latter. While more data is necessary to help clarify these patterns and formulate a more comprehensive analysis, I have provided a stepping off point for which to analyze the vowel alternations in Northern Dagaare number marking.

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